AIRI Team PITB

AI Internship - Task 1

Helmet Detection System Using YOLOv8

End-to-End Computer Vision Project Report

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Program	AIRI Team PITB - AI Internship, Task 1
Submitted to	Omar Farooq, Associate AI/ML Engineer
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Model / Platform	YOLOv8n (Ultralytics) on Google Colab, Tesla T4 GPU
Classes	helmet, no_helmet, person

Executive Summary

This project builds a complete object-detection pipeline that detects helmets, bare heads (no helmet) and persons. I collected and manually reviewed images, annotated 157 of them in Roboflow, prepared a YOLO-format dataset, trained a YOLOv8n model on Google Colab, evaluated it on validation and test data, ran inference on unseen images and analysed the errors.

Images used	Train / Val / Test	Boxes	Test mAP@0.5	Val mAP@0.5
157	109 / 31 / 17	364	0.630	0.553

Contents: 1 Problem Definition, 2 Data Collection, 3 Annotation, 4 Dataset Organisation, 5 Training, 6 Evaluation, 7 Inference, 8 Error Analysis, 9 Conclusion.

1. Problem Definition

This project builds an object-detection system that finds **helmets**, **people without helmets** and **persons** in images. It can support basic safety monitoring on roads and workplaces by flagging riders or workers who are not wearing helmets. The final output is an image or video with bounding boxes, class labels and confidence scores.

ID	Class	Meaning
0	helmet	A visible safety helmet on a head
1	no_helmet	A visible head without a helmet
2	person	The full visible body of a person

2. Data Collection

Images were taken from a public Roboflow Universe dataset named **helmet detection** (universe.roboflow.com/project-pdvie/helmet-srsz5). I uploaded 176 images into my own Roboflow project and reviewed them manually. Duplicate or near-identical images were removed, and 157 images were annotated and used for training and evaluation. The images show different angles, backgrounds and people (motorcycle riders, cyclists, small groups).

3. Data Annotation

Annotation was done in **Roboflow Annotate** with bounding boxes, and exported in **YOLOv8 format**. Boxes were drawn tightly: a helmet box covers only the helmet, a no_helmet box covers the bare head, and a person box covers the whole body. Only clearly visible objects were labelled. All label files were checked automatically and contained **0 invalid lines**.

Item	Value
Images annotated and used	157
Total boxes	364 (helmet 125, no_helmet 57, person 182)
Format	YOLO: class_id x_center y_center width height (normalised 0-1)
Preprocessing	Auto-orient, resize 640x640 (stretch)
Augmentation	None (to avoid duplicate images)

4. Dataset Organisation and data.yaml

The 157 images were randomly split (seed 42) into **109 train (70%)**, **31 validation (20%)** and **17 test (10%)** images, each with a matching .txt label file, following the required folder structure.

data.yaml explained: **path** is the dataset root folder; **train** are the images the model learns from; **val** are images used to check progress during training; **test** are held-out images for the final score; **names** maps class IDs to class names (0 helmet, 1 no_helmet, 2 person).

5. Training Configuration

Setting	Value
Model	YOLOv8n (pretrained yolov8n.pt)
Epochs	30
Image size	640
Batch size	8
Platform / GPU	Google Colab, Tesla T4
Training time	1.9 minutes
Best weights	models/best.pt
Training outputs	outputs/training_results/yolov8n_baseline

6. Evaluation

The model was evaluated on the validation set (31 images, 80 objects) and the test set (17 images, 34 objects).

Split	Precision	Recall	mAP@0.5	mAP@0.5:0.95
Validation	0.598	0.496	0.553	0.232
Test	0.585	0.586	0.630	0.312

Per-class results (mAP@0.5)

Class	Val P	Val R	Val mAP50	Test P	Test R	Test mAP50
helmet	0.668	0.577	0.639	0.693	0.357	0.570
no_helmet	0.439	0.286	0.366	0.420	0.667	0.607
person	0.688	0.625	0.654	0.641	0.735	0.714

Metric meanings. Precision: of all objects the model detected, how many were correct. **Recall**: of all real objects in the images, how many the model found. **mAP** (mean Average Precision): the average detection quality over all classes; mAP@0.5 counts a box as correct if it overlaps the true box by at least 50%, and mAP@0.5:0.95 averages over stricter overlap levels from 50% to 95%.

Best class: person (highest mAP@0.5 on both splits: 0.654 val, 0.714 test). It has the most examples (182 boxes) and is usually the largest object in the image. **Weakest class:** no_helmet on validation (mAP@0.5 0.366, recall 0.286), with fewer examples (57 boxes) and an appearance that is easily confused with helmet-wearing heads. Note that the test set contains only 3 no_helmet objects, so its test number is not very reliable. On the test set, helmet recall (0.357) is the lowest value.

Training curves. Figure 1 shows the training and validation losses and the metrics over the 30 epochs.

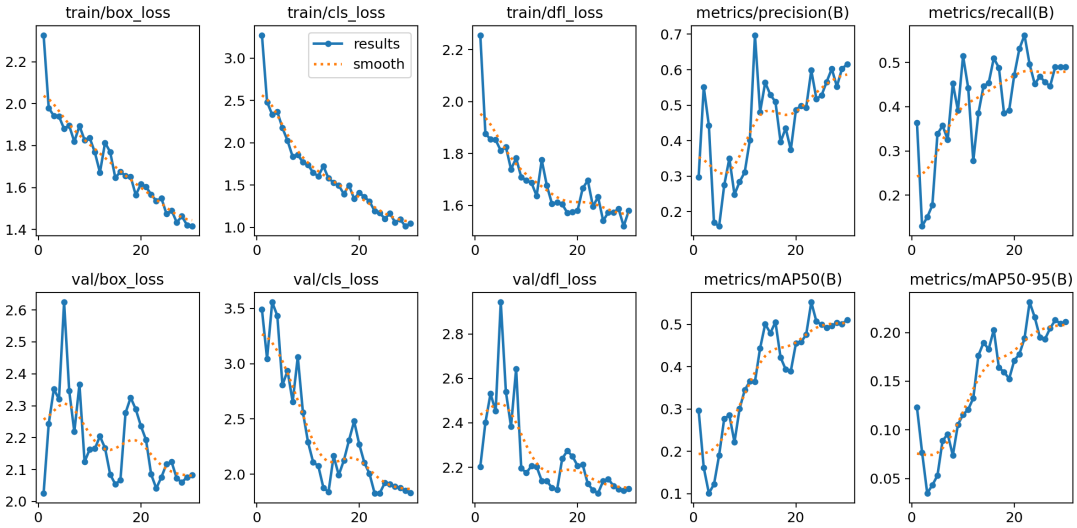


Figure 1: Training/validation loss curves and metrics (results.png).

Training box, class and DFL losses fall steadily, and validation losses also fall (with noise, because the validation set has only 31 images). Precision, recall and mAP rise over the epochs and the best mAP@0.5 (about 0.55) is reached near epoch 23. The curves have not fully flattened, so longer training or more data could still help.

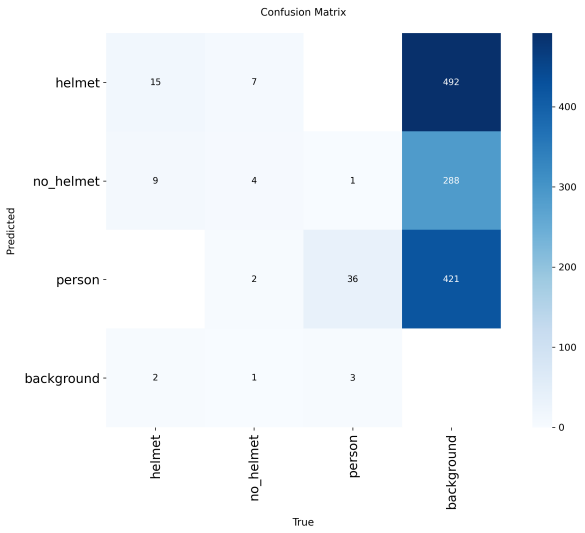


Figure 2: Confusion matrix on the validation set (confusion_matrix.png).

Reading the confusion matrix. Columns are the true class and rows the predicted class. Of the 26 true helmets, 15 were predicted correctly, 9 were predicted as no_helmet and 2 were missed. Of the 14 true no_helmet objects, 4 were correct, 7 were predicted as helmet, 2 as person and 1 was missed. Of the 40 persons, 36 were correct. The main confusion is therefore between helmet and no_helmet. The large background counts in the matrix come from low-confidence boxes that the validation step keeps at its default low confidence threshold; they do not appear at the 0.35 threshold used for inference.

Why mistakes happen: mainly confusion between helmet and no_helmet (see the confusion matrix), plus a small training set (109 training images), a small model (YOLOv8n), few no_helmet examples, and likely small or partly hidden heads in group photos.

7. Inference on Unseen Images

The final model was run on the 17 held-out test images with confidence threshold 0.35. All prediction images show bounding boxes, predicted labels and confidence scores and are saved in *outputs/predictions/test_predictions* (17 images, above the required 15).

8. Error Analysis

An automatic comparison of the number of objects of each class per image, predicted versus ground truth, on the 17 test images (confidence 0.35) found **14 images with at least one difference**. Ten of them are listed below (image names are shortened). The reasons are likely explanations (hypotheses), not confirmed causes.

Image	Actual	Model prediction	Error type	Possible reason
BikesHelmets136	helmet, person	person	False negative	Helmet small or partly hidden
BikesHelmets17	helmet, person	person	False negative	Helmet missed, few similar training examples
BikesHelmets246	helmet, person	person	False negative	Unusual helmet colour or shape
BikesHelmets36	helmet, person	no_helmet, person	Wrong class	Helmet and bare head look similar at low resolution
BikesHelmets184	helmet, person	nothing detected	False negative	Small or distant objects
BikesHelmets405	helmet, person	nothing detected	False negative	Small or distant objects
BikesHelmets201	no_helmet, person	nothing detected	False negative	Few no_helmet training examples
BikesHelmets242	helmet, person	person, person	False negative + false positive	Helmet missed, one extra person box
BikesHelmets169	helmet, person	helmet, helmet, person	False positive	Extra helmet box, possibly an unlabelled helmet
BikesHelmets133	no_helmet, person	no_helmet	False negative	Person partly hidden by another rider

Across the 14 images, the most common error is a missed helmet (9 images), followed by a missed person (6 images); there was one wrong-class case and a few extra boxes (false positives). The validation confusion matrix shows that helmet and no_helmet are also frequently confused with each other (9 helmets predicted as no_helmet and 7 no_helmet predicted as helmet). Small objects, partial occlusion and similar-looking classes are the likely causes; some extra boxes may also come from annotation gaps in the ground truth.

Suggested improvements

1. Add more images, especially small or distant helmets and no_helmet examples, to balance the classes.
2. Train longer or use a larger model such as YOLOv8s.
3. Re-check annotations for missing or loose boxes.
4. Add more low-light, blurry and crowded examples.

9. Conclusion and Future Work

I completed the full workflow: problem selection, data collection, annotation, YOLO dataset preparation, training on Google Colab, evaluation, inference and error analysis. The baseline YOLOv8n model reached mAP@0.5 of 0.553 (validation) and 0.630 (test). Results can be improved with more and more balanced data and a larger model. Future work: deploy a Streamlit or Hugging Face demo and test on real-time video.

10. Project Deliverables

Deliverable	Location in project folder
Google Colab notebook with outputs	notebooks/training_notebook.ipynb
Annotated dataset (YOLO format) and data.yaml	dataset/images, dataset/labels, dataset/data.yaml
Trained model	models/best.pt
Training results, loss curves, confusion matrix	outputs/training_results/yolov8n_baseline
Prediction images (17 test images)	outputs/predictions/test_predictions
Error-analysis table and candidate list	Section 8 of this report, outputs/error_candidates.csv
README and requirements	README.md, requirements.txt
Final report	report/final_report.pdf

Dataset credit: Roboflow Universe, project 'helmet detection' (project-pdvie/helmet-srsz5). **Tools:** Python, Google Colab, Google Drive, YOLOv8, Ultralytics, OpenCV, Roboflow, Matplotlib.